

From Clinical Data Warehouses to Epidemiological Indicators: A Methodology for Real-Time Surveillance

Thomas JAYLET^a, Amélie BREJOT^a, Valérie BERTAUD^a, Marc CUGGIA^a,
Emmanuelle SYLVESTRE^{a,b} and Morgane PIERRE-JEAN^{a,1}

^a *Université de Rennes, CHU Rennes, Inserm LTSI-UMR 1099, 35000 Rennes, France*

^b *CHU Martinique, Centre de Données Cliniques, Martinique, France*

ORCID ID: Morgane PIERRE-JEAN <https://orcid.org/0000-0002-9133-780X>

Abstract. Clinical data warehouses (CDWs) provide a robust foundation for generating real-world epidemiological indicators from electronic health records. By exploiting routinely collected hospital data, these infrastructures enable near real-time monitoring of infectious diseases. Their value increases when complemented by external data streams, including web search queries or meteorological variables. Evidence from recent studies shows that combining these heterogeneous signals with statistical models, natural language processing (NLP), and machine learning (ML) enhances both the timeliness and accuracy of outbreak detection and forecasting. The project aims to combine novel methods and apply them to our local CDW by considering the changes in open data.

Keywords. Clinical data warehouse, indicators, epidemiology

1. Introduction

In France, traditional flu surveillance relies on the Sentinelles system, which provides valuable but delayed indicators (1-3 weeks). Clinical Data Warehouses (CDWs) can reduce this lag by using structured and unstructured hospital data to deliver timely, validated case counts enriched with clinical details. Studies have shown that CDW-derived indicators correlate well with influenza incidence and may outperform digital sources alone [1]. Combining CDWs with external data, such as weather variables, further improves predictive accuracy [2]. CDWs thus offer a robust clinical foundation, while digital and environmental data enhance real-time responsiveness.

2. Combining data extraction and prediction models

The strength of CDWs lies in their ability to combine structured data with unstructured clinical text. A significant proportion of relevant variables can be found in unstructured notes, making Natural Language Processing (NLP) essential for extracting disease mentions or functional indicators. Research has demonstrated the effectiveness of NLP

¹ Corresponding Author: Morgane PIERRE-JEAN, morgane.pierre-jean@chu-rennes.fr.

in recovering symptoms [3]. Given these considerations, the proposed methodology builds on these advantages by extracting clinical data (e.g., diagnoses and symptoms) from the hospital eHOP CDW using structured queries alongside text-based searches within unstructured clinical notes. In combination with external data sources, these data serve as inputs to Machine Learning (ML) approaches, including Random Forests (RF) or Support Vector Machines (SVMs), capturing complex temporal patterns in heterogeneous datasets [4]. While ML models often outperform classical statistical and time-series approaches, their performance remains dependent on data quality.

3. Discussion and Conclusion

We propose combining heterogeneous data sources through ML models to consistently improve forecasting accuracy and responsiveness of epidemiological indicators. Initial ML models for influenza epidemic prediction were built using penalized logistic regressions (Lasso, Ridge, ElasticNet), Random Forests and SVMs, based on weekly aggregated data from eHOP CDW, the Sentinelles system, and meteorological sources for binary classification of epidemic status. A total of 18 models were trained and evaluated against historical epidemic periods reported by Santé publique France, of which 11 achieved F1 scores ranging from 78.8% to 88.2%, and 9 were able to predict the start of the 2024-2025 influenza epidemic in time.

Acknowledgments

This work was carried out within the framework of the Consortium Creation of a national network of hospitals involved in the epidemiological surveillance and emergency response (Orchidée). This information is available on the website: <https://www.santepubliquefrance.fr/en/orchidee-project-developing-a-new-national-epidemiological-surveillance-hospital-network-through-secondary-health-data-usage-in-hospitals>.

References

- [1] Sylvestre E, Cécilia-Joseph E, Bouzillé G, Najioullah F, Etienne M, Malouines F, et al. The Role of Heterogenous Real-world Data for Dengue Surveillance in Martinique: Observational Retrospective Study. *JMIR Public Health and Surveillance*. 2022 Dec 22;8(12):e37122. doi: 10.2196/37122.
- [2] Poirier C, Hswen Y, Bouzillé G, Cuggia M, Lavenu A, Brownstein JS, et al. Influenza forecasting for French regions combining EHR, web and climatic data sources with a machine learning ensemble approach. *PLOS ONE*. 2021 May 19;16(5):e0250890. doi: 10.1371/journal.pone.0250890.
- [3] Petit-Jean T, Gérardin C, Berthelot E, Chatellier G, Frank M, Tannier X, et al. Collaborative and privacy-enhancing workflows on a clinical data warehouse: an example developing natural language processing pipelines to detect medical conditions. *J Am Med Inform Assoc*. 2024 May 20;31(6):1280–90. doi: 10.1093/jamia/ocae069.
- [4] Poirier C, Bouzillé G, Bertaud V, Cuggia M, Santillana M, Lavenu A. Gastroenteritis Forecasting Assessing the Use of Web and Electronic Health Record Data With a Linear and a Nonlinear Approach: Comparison Study. *JMIR Public Health and Surveillance*. 2023 Jan 31;9:e34982. doi: 10.2196/34982.